

(C++) Computer Programming

Assignment - 4

1. For each of the following programs, point out the error, if any. If there is no error, determine the output of the programs.

a)

```
int main()
{
    int x = 15;

    for (x = 10; x >= 10; x--)
    {
        cout << x << endl;
    }
}
```

c)

```
int main()
{
    int K = 1;
    int sum = 0;
    int count = 0;
    while (K != 10)
    {
        sum++;
        count++;
        K++;
    }
    cout << "sum = " << sum << endl;
    cout << "count = " << count << endl;
    return 0;
}
```

e)

```
int main()
{
    int m = 1, n = 5;

    for (; m + n < 10; m--)
    {
        cout << "Hello" << endl;
        n = n + 2;
    }
    return 0;
}
```

b)

```
int main()
{
    int total = 0;
    int x = 0;

    do
    {
        total = total + x;
    }
    while (total != 10);
}
```

d)

```
int main()
{
    int i, j, k = 0;

    for (i = 0; i < 4; i++)
    {
        for (j = 0; j < i; ++j)
        {
            k += i + j - 1;
        }
    }

    cout << k << endl;
    return 0;
}
```

f)

```
int main()
{
    int i;

    for (i = 0; i < 10; )
    {
        cout << i << endl;
    }
    return 0;
}
```

g)

```
int main()
{
    int x = 10;

    do
    {
        cout << x << endl;
        x = x / 2;
    }
    while (x > 0);

    return 0;
}
```

h)

```
int main()
{
    int i = 1;

    for (;;)
    {
        cout << i++ << endl;

        if (i > 10)
            break;
    }

    return 0;
}
```

2. Write a program to find the minimum number of notes required for a given amount. The available notes are Rs. 1, Rs. 2, Rs. 5, Rs. 10, Rs. 20, Rs. 100, and Rs. 500.
3. Write a program to count the number of digits in a given integer.
4. Write a program to check whether a given number is a palindrome or not.
5. Write a program to loop over $0 \leq i, j < 10$ and print the first pair (i, j) for which $i \times j > N$, where N is a number entered by the user. Test the program with $N = 40$.
6. Write a C++ program to find the approximate value of the following series, where x and n (number of terms) are entered by the user.

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots$$

7. Write a program to print the following patterns:

a)

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

b)

```
 *
 * *
 * * *
 * *
 *
```

8. Write a program to find the GCD (Greatest Common Divisor) of two given numbers.